Equal forces can have unequal effects

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Force pair / ES: par de fuerzas / PT: par de forças

A-on-B and B-on-A: equal and opposite.

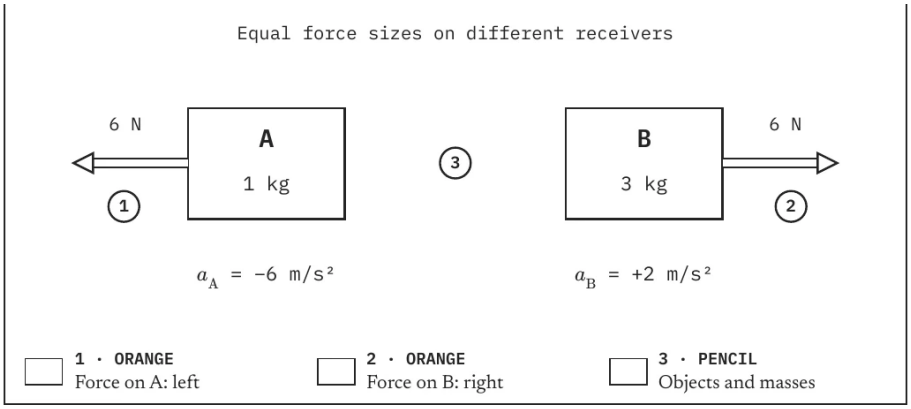
Also on your note page: Receivers, Mass effect, System.

B READ THE RELATIONSHIP

F(A on B) = −F(B on A); each receiver has its own a = F/m

As you read, find: on A · on B · same strength.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Cart A has mass 1 kg ; cart B has mass 3 kg . Their interaction pushes A left and B right with 6 N each. Find both accelerations.

Given m A = 1 kg m B = 3 kg right is +

Rule F = ma on each cart. No other horizontal forces.

Work a A = −6/1 = −6 m/s² a B = +6/3 = +2 m/s²

Answer −6; +2 m/s² A has 3 times B's acceleration size.

E SAVE FOR THE FINAL MEETING

Earth and fictional XJ pull on each other equally at the same instant. Their very different masses give different accelerations; the pair does not cancel in XJ's own force diagram.

Read the physics: Force pair

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

If A pushes B, B pushes A at the same instant. The forces have the same strength. Swap the names to find the other receiver.

02 HOW IT WORKS

Draw one diagram for each receiver. Put only the force on that receiver in its diagram. These two arrows cannot cancel on one object.

03 WHAT TO CHECK

For the same force size, the smaller mass has the larger acceleration. Balanced forces on one object are a different idea.

04 TRY THE EXAMPLE FROM YOUR NOTES

Cart A has mass 1 kg ; cart B has mass 3 kg . Their interaction pushes A left and B right with 6 N each. Find both accelerations. −6; +2 m/s² A has 3 times B’s acceleration size.

Model check: F(A on B) = −F(B on A); each receiver has its own a = F/m.

05 CONNECT TO XJ

Earth and fictional XJ pull on each other equally at the same instant. Their very different masses give different accelerations; the pair does not cancel in XJ's own force diagram.

SAY IT BEFORE YOU WRITE

Do your two arrows end up on the same object? ___ pushes on ___. Its partner is ___ pushing on ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Force pair

ES: par de fuerzas / PT: par de forças

Meaning in my words	How can you tell?
How would you explain force pair to a friend? In my words, this means ___.	Earth pulls an apple down. What is the partner force, and which object feels it? I can tell because ___.
Example or sketch	**Non-example and why**
How would you label the pair when a hand pushes a cart? My example shows ___.	Are the table's push and Earth's pull on one book a third-law pair? I would change ___ because ___.

Receivers

ES: objetos que reciben las fuerzas / PT: objetos que recebem as forças

Meaning in my words	How can you tell?
In an interaction pair, what are the two receivers? In my words, this means ___.	In "the wall pushes on my hand," where does that force arrow belong? I can tell because ___.
Example or sketch	**Non-example and why**
When my hand pushes on the wall, which object receives that other force? My example shows ___.	Can I draw both contact forces of that pair on the hand? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Mass effect

ES: efecto de la masa / PT: efeito da massa

Meaning in my words	How can you tell?
How does a cart's mass affect its response to a push? In my words, this means ___.	Two skaters push off with equal net force magnitudes. Which skater has greater acceleration? I can tell because ___.
Example or sketch	**Non-example and why**
A 10 N net force acts separately on 2 kg and 5 kg carts. Compare their accelerations. My example shows ___.	The lighter cart accelerates more. Does that prove it felt the larger interaction force? I would change ___ because ___.

System

ES: sistema / PT: sistema

Meaning in my words	How can you tell?
How would you explain system to a friend? In my words, this means ___.	Draw one boundary around two interacting carts. Where are their bumper forces relative to it? I can tell because ___.
Example or sketch	**Non-example and why**
If a hand pushes one cart from outside that boundary, where does its force go? My example shows ___.	When you choose just one cart, can you still drop the force from the other cart? I would change ___ because ___.

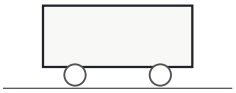
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

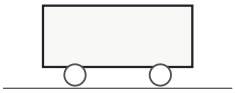
Label both receivers first. Draw equal force arrows on different objects, pointing toward one another for gravity.

Labels you can use: on A · on B · same strength.

Receiver A



Receiver B



Tell a partner

___ pushes on ___. Its partner is ___ pushing on ___.

Two pulls. Two receivers.

14

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Place an interaction pair on its two receivers.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

Put each gravitational arrow on the body that receives it. Send the animator two corrected object panels.

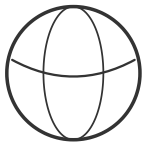
SCIENCE GRAPHICS PRODUCER ☐

Write a split-screen caption naming who pulls whom. Explain why equal force does not require equal acceleration.

Word help: interaction / receiver / equal and opposite

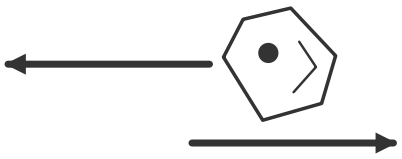
Earth pulls on XJ. A public animation shows both gravitational arrows on XJ, with the caption “they cancel.”

EARTH PANEL



Earth

XJ PANEL



PROPOSAL: “They cancel.”

01 OUR FIRST CALL

Do the two arrows cancel on XJ? Explain using your corrected panels.

___ pulls on ___. That arrow belongs on ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Which arrow belongs in the updated shot? Name its source and receiver.

Now we would say ___ because ___.

Switch speakers. If Earth and XJ are one combined system, which forces in this interaction are internal to that system?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks